

Unified Dual-Mask Physical Model for Non-Lambertian Light Field Depth Estimation

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Abstract—Light field depth estimation is fundamentally challenged by non-Lambertian surfaces, where specular and scattering reflections violate the epipolar plane image (EPI) linearity assumption. Existing methods heavily rely on high-resolution angular spectra or uniform EPI constraints, leading to severe performance degradation under low angular resolution and complex reflectance. To address this, we propose a unified dual-mask physical model that decouples illumination, depth, and material properties through a novel three-layer angular signal decomposition. Specifically, we model the angular signal as a superposition of a direct current component, linear disparity, and a material residual. Instead of relying on fragile frequency features, we extract robust geometric features, such as angular gradients, to train a random forest classifier for precise Lambertian and non-Lambertian pixel segregation. These physical priors guide a GeometricDualMask architecture, which adaptively routes pixels to specialized EPI and geometric processing branches via a dual-mask mechanism, optimized with domain-balanced sampling. Extensive evaluations across three datasets demonstrate that our method achieves an overall mean absolute error (MAE) of 0.133 and a mixed/urban scene MAE of 0.081, significantly outperforming state-of-the-art baselines. Furthermore, the material classifier attains an 89.6

Index Terms—Light field imaging, depth estimation, non-Lambertian reflectance, angular signal decomposition, epipolar plane image, dual-mask routing.

I. Introduction

LIGHT field imaging captures both spatial and angular information, enabling precise 3D reconstruction and depth estimation [?]. Accurate depth estimation from light fields serves as a core component for advanced applications in autonomous driving, augmented reality, and robotic manipulation [?]. Most existing light field depth estimation methods rely heavily on Epipolar Plane Images (EPIs) and the Lambertian reflectance assumption [?]. Under this assumption, a surface point exhibits consistent radiance across different viewpoints, forming clear linear structures in the EPI space.

However, real-world scenes frequently contain non-Lambertian surfaces and complex textures, which severely violate the Lambertian assumption. Specular reflections and volumetric scattering alter pixel intensities across viewpoints, destroying the linear EPI geometry and generating X-shaped patterns instead of straight lines [?]. Similarly, complex textures cause severe slope ambiguity

in EPIs, obscuring the true depth cues. Consequently, state-of-the-art EPI-based networks experience significant performance degradation in these challenging regions. For instance, on standard benchmarks, the Mean Absolute Error (MAE) for non-Lambertian regions reaches 0.411, substantially exceeding the acceptable threshold of 0.25. Meanwhile, the MAE for complex-textured Lambertian regions degrades to 0.387, failing to meet the target of 0.16.

The severity of this problem is further compounded by extreme data scarcity. Current non-Lambertian light field datasets provide only a handful of training scenes. For example, some benchmarks contain merely four specific scenarios, which remains substantially below the minimum requirement for data-driven generalization [?]. To address these failures, previous studies have explored deeper architectures, attention mechanisms, and complex loss functions [?]. Nevertheless, extensive empirical evaluations involving 132 experiments across 16 major research directions demonstrate a key finding. Purely architectural modifications cannot overcome the physical violation of EPI assumptions. The performance ceiling of single-branch EPI networks is strictly bounded by the physical properties of the surfaces rather than the network capacity.

Therefore, it is urgently necessary to revisit the physical formation of light field signals. We must decouple the intertwined factors of illumination, depth, and material reflectance to resolve this bottleneck. Relying solely on high-resolution angular frequency spectrums is impractical for commercial light field cameras with low angular resolutions, such as 9×9 views [?]. A unified physical model that explicitly identifies and routes non-Lambertian pixels is essential to overcome the inherent limitations of EPI-based depth estimation. By establishing a robust material prior and applying adaptive physical constraints, we can effectively mitigate the EPI assumption conflicts. This approach enables reliable depth estimation across diverse and complex real-world scenes.

Traditional light field depth estimation methods primarily rely on handcrafted epipolar plane image (EPI) features or high-frequency angular spectra [?]. Methodologically, these approaches strictly assume Lambertian reflectance, rendering them highly vulnerable to specular highlights and volumetric scattering. When non-Lambertian surfaces introduce X-shaped patterns in EPIs, the assumed linear geometric constraints collapse entirely. Technically, many traditional algorithms depend on high-resolution angular

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spectra, such as 2D discrete Fourier transforms. However, these spectral features severely degrade under low angular resolutions, such as the standard 9×9 angular grid of commercial cameras. Furthermore, complex textures induce severe slope ambiguity, causing traditional gradient-based operators to fail in distinguishing true depth cues from surface albedo variations [?].

To overcome the limitations of handcrafted features, early deep learning methods employ convolutional neural networks to extract EPI representations implicitly [?]. Methodologically, these models treat depth estimation as a purely data-driven regression task, largely ignoring the underlying physical imaging process. Consequently, they exhibit poor generalization when the core EPI assumption is violated. Technically, these early architectures utilize a single unified network to process all surface types. This design inevitably causes feature collision between Lambertian and non-Lambertian pixels. The network struggles to learn a shared representation that simultaneously preserves the linear EPI structures for matte surfaces and captures the complex angular variations for glossy regions [?]. As a result, the depth accuracy for non-Lambertian regions remains substantially below acceptable thresholds.

Recent advanced approaches attempt to mitigate these issues by introducing attention mechanisms, dual-branch architectures, or defocus-based auxiliary heads [?]. Methodologically, however, these structural enhancements fail to address the root causes of performance degradation. Our extensive analysis demonstrates that the primary bottlenecks stem from physical assumption violations and extreme data scarcity, rather than insufficient network capacity or suboptimal hyperparameters. Blindly increasing model complexity leads to inefficient computational consumption without resolving the underlying physical conflicts. Technically, current methods lack explicit material priors and physical constraints. Existing domain-specific masking or branching mechanisms are typically learned implicitly. Under extreme data scarcity, where non-Lambertian datasets provide merely four training scenes, these implicit strategies easily overfit and fail to achieve robust cross-domain generalization [?].

In summary, existing methods either rely on invalid physical assumptions for non-Lambertian surfaces or employ overly complex networks without addressing the root causes of EPI failure. Therefore, there is an urgent need for a unified approach that explicitly decouples illumination, depth, and material from a physical perspective, and adaptively routes different reflectance properties within a single architecture.

In this paper, we propose a unified dual-mask physical model, named GeometricDualMask, to address the inherent limitations of epipolar plane image (EPI) assumptions in non-Lambertian light field depth estimation. Instead of relying on high-resolution angular spectra that degrade under low angular sampling, we introduce a three-layer angular signal decomposition mechanism. This physical model decouples the angular signal into ambient illumination, parallax depth, and material reflectance components.

By extracting geometric features such as angular gradients from the residual component, we generate pixel-level dual masks to route Lambertian and non-Lambertian pixels into specialized processing branches. This architecture effectively resolves the feature collision problem in mixed scenes, enabling adaptive depth estimation under complex reflectance properties.

The main contributions of this paper are summarized as follows:

- **Three-Layer Angular Signal Decomposition and Geometric Material Classification:** We propose a physical decomposition model, formulated as $I(u, v) = DC + a \cdot u + b \cdot v + \varepsilon(u, v)$, to decouple illumination, depth, and material reflectance. By extracting geometric features like angular gradients from the residual term $\varepsilon(u, v)$ and training a random forest classifier, we achieve accurate binary classification of Lambertian and non-Lambertian pixels. This mechanism resolves the failure of spectral features under low angular resolutions and provides robust material priors for subsequent networks.
- **Geometry-Guided Dual-Mask Routing Architecture:** We design the GeometricDualMask network, which utilizes the extracted geometric priors to generate dual masks for adaptive pixel routing. Lambertian pixels are processed by an EPI branch incorporating planar regularized sampling operators, while non-Lambertian pixels are handled by a geometric branch detecting X-shaped patterns. Combined with domain-balanced sampling and a mixed training strategy, this unified architecture substantially improves depth estimation accuracy in mixed and complex reflective scenes.
- **Physical Boundary and Root Cause Analysis of EPI Assumption Failure:** We systematically quantify the physical mechanisms causing EPI assumption failures on non-Lambertian and complex-textured Lambertian surfaces. Through 132 extensive experiments, we demonstrate that the violation of physical assumptions and data scarcity, rather than network structures, are the root causes of performance degradation. This analysis establishes the theoretical performance boundaries of EPI-based methods and provides clear guidance for future non-EPI approaches and dataset construction.

Trained with an initial learning rate of 10^{-4} , a batch size of 2, and 8 gradient accumulation steps, our GeometricDualMask achieves an overall mean absolute error (MAE) of 0.133 and a mixed urban scene MAE of 0.081 on multiple light field datasets. These results substantially outperform existing single-branch baselines, validating the efficacy of the dual-mask physical model in mitigating EPI assumption conflicts and enhancing depth estimation robustness across diverse surface types. To contextualize our contributions within the broader research landscape, we begin by reviewing the traditional methodologies that have historically guided light field depth estimation.

II. Related Work

A. Traditional Light Field Depth Estimation Methods

Many traditional depth estimation methods for light field (LF) cameras have been proposed, primarily leveraging physical cues such as Epipolar Plane Image (EPI) slopes, defocus blur, and stereo matching [?]. These approaches substantially rely on geometric modeling under the strict Lambertian reflectance assumption, where a surface point emits consistent radiance across all viewpoints, forming distinct linear structures in the EPI space.

Wanner et al. [?] pioneered the use of EPI structural analysis by deriving a theory that links the local orientation of EPI lines to scene depth. Specifically, they employed a structure tensor to calculate the dominant slope of EPI lines, effectively resolving the ambiguity in textured regions. While this approach achieves notable accuracy on Lambertian surfaces, it intrinsically assumes that EPI lines are perfectly straight. Consequently, when encountering non-Lambertian surfaces such as specular reflections, the EPI lines distort into X-shaped patterns, causing the structure tensor-based slope calculation to fail completely.

To mitigate the limitations of purely slope-based methods, Tao et al. [?] extended the physical modeling by combining defocus and correspondence cues. They formulated a depth estimation framework that leverages the depth-dependent blur in focal stacks and the angular shift in sub-aperture images. By fusing these two complementary cues, their method effectively improves depth accuracy in weakly textured regions. However, this fusion strategy heavily relies on the consistency between defocus and correspondence cues. In the presence of glossy or mirror-like surfaces, the specular highlights shift inconsistently across viewpoints, leading to severe conflicts between the two cues and causing the algorithm to hallucinate their depths.

In contrast to focal stack-based approaches, stereo matching paradigms focus on leveraging multi-view geometry directly from sub-aperture images. Jeon et al. [?] proposed an adaptive cost volume method that incorporates a phase-shift technique to compensate for sub-pixel displacements. This approach effectively refines depth boundaries and resolves the ambiguity in fine structures. Nonetheless, it primarily depends on the photo-consistency assumption across viewpoints. When applied to spatially-varying (SV) reflectance or transparent or mirror (ToM) surfaces, the matching cost becomes under-constrained, leading to severe mismatches in non-Lambertian regions.

Despite their extensive contributions, these traditional approaches share inherent deficiencies that restrict their applicability in real-world scenarios. On the one hand, they heavily rely on hand-crafted features and the strict Lambertian assumption. When confronting non-Lambertian materials (e.g., specular reflections and highlights), the EPI geometry degenerates from straight lines to X-shaped patterns, rendering the slope and matching calculations entirely invalid. On the other hand, they lack the physical-

level decoupling capability among illumination, depth, and material reflectance. As a result, they exhibit significantly poor robustness in complex textured and mixed-reflection scenes, failing to provide reliable priors for modern complex scenarios. These inherent limitations not only provide the motivation for our subsequent quantitative analysis of the physical mechanisms behind EPI failure but also underscore the necessity of explicitly decoupling geometric and material properties. To this end, our proposed unified dual-mask physical model addresses these gaps by introducing a three-layer angular signal decomposition and a geometric dual-mask routing mechanism, enabling adaptive processing of complex reflectance characteristics without being constrained by the traditional EPI assumptions.

B. Deep Learning-based General Epipolar Depth Estimation Methods

Following the limitations of handcrafted physical cues, deep learning-based methods have emerged to implicitly learn the mapping from Epipolar Plane Images (EPIs) to depth maps. Shin et al. [?] pioneered this direction by proposing EPINET, a fully convolutional network that extracts both spatial and angular features from EPIs and the central view. By concatenating these multi-dimensional features, the network effectively regresses depth values, achieving considerable accuracy on standard Lambertian benchmarks. Subsequent improvements [?] introduced multi-scale EPI processing to enhance local geometric representations. However, the reliance on standard convolutional kernels inherently restricts the receptive field along the epipolar lines, making it challenging to decouple and aggregate long-range geometric dependencies in high-resolution light fields.

To overcome the restricted receptive field and mitigate feature interference, recent studies have introduced attention mechanisms and feature disentanglement strategies. For instance, Wang et al. proposed a disentangled feature network that explicitly separates angular and spatial representations. By employing spatially-varying (SV) attention modules, this approach adaptively aggregates multi-view cues, significantly enhancing depth estimation in texture-less and occluded regions. Similarly, Meng et al. [?] leveraged an adaptive cost volume to regularize the distribution of epipolar features, further improving robustness against local ambiguities. While these attention-driven architectures successfully expand the effective receptive field and refine feature aggregation, they primarily focus on optimizing the network capacity without reconsidering the underlying physical assumptions of the EPI structure.

Further pushing the boundaries of receptive field expansion, Transformer-based architectures have been explored to capture global epipolar dependencies. Liang et al. [?] introduced a light field Transformer that applies self-attention mechanisms specifically along the epipolar lines. This design allows the model to dynamically weigh distant

pixels within the same EPI, effectively resolving the aliasing bimodal distribution and slope ambiguities in complex textures. Consequently, empirical evaluations on public datasets demonstrate that such Transformer-based models achieve state-of-the-art performance. Nonetheless, the dramatic increase in computational complexity and the implicit reliance on the linear EPI prior remain critical bottlenecks when generalizing to real-world scenarios with diverse material properties.

Despite these extensive architectural advancements, general deep learning-based EPI-based methods share inherent physical limitations. On the one hand, the vast majority of these networks implicitly assume that EPIs possess perfect linear structures. When the input contains non-Lambertian pixels, the networks cannot distinguish whether the EPI slope blur is caused by continuous depth variations or material reflections, leading to a dramatic performance degradation in non-Lambertian regions. On the other hand, these methods lack explicit physical constraints and material awareness. They attempt to fit the data merely by increasing network complexity (e.g., deeper networks or attention mechanisms), which cannot overcome the physical performance boundary caused by the violation of the EPI assumption. This proves that pure parameter tuning cannot resolve physical assumption conflicts. In contrast, our proposed unified dual-mask physical model explicitly addresses these physical conflicts by formulating a three-layer angular signal decomposition and a geometric dual-mask routing mechanism. This design inherently decouples geometric and material properties, enabling adaptive processing of complex reflectance characteristics without being constrained by the implicit linear EPI assumptions of existing deep networks.

C. Summary

In summary, both traditional and deep learning-based light field depth estimation methods exhibit inherent limitations when handling non-Lambertian surfaces. Traditional methods heavily depend on hand-crafted features and strict Lambertian assumptions, failing completely when EPI geometries degenerate into X-shaped patterns. Conversely, deep learning-based approaches attempt to overcome these issues through increased network complexity, attention mechanisms, and global receptive fields. However, they still implicitly rely on linear EPI priors and lack explicit physical constraints, leading to suboptimal generalization and significant performance degradation in mixed-reflection scenes. Crucially, neither paradigm effectively decouples illumination, depth, and material reflectance at the physical level. To bridge this critical gap, this paper proposes a unified dual-mask physical model that explicitly formulates the physical mechanisms of EPI degradation and adaptively routes angular signals, thereby achieving robust and accurate depth estimation in complex real-world scenarios. Building upon this conceptual foundation, we now detail the complete technical pipeline, mathematical formulations, and implementation specifics of the proposed framework.

III. Methodology

In this section, we present the overall architecture of the proposed GeometricDualMask model, which aims to address the challenge of accurate depth estimation in light fields containing non-Lambertian surfaces. Different from previous works that heavily rely on the strict Lambertian assumption and high-resolution angular spectrum analysis, our framework explicitly models the physical boundary of Epipolar Plane Image (EPI) failure. As illustrated in Fig. 2, the proposed unified architecture incorporates four core components: a Three-Layer Angular Signal Decomposition module, a Lambertian EPI Branch, a Non-Lambertian Geometric Branch, and a Dual-Mask Weighted Fusion module. This design enables the adaptive routing of pixels to specialized processing branches based on their reflectance properties, thereby maintaining robust performance across diverse material domains.

The input to our model is a light field image sequence captured with a 9×9 angular sampling grid. Let $I(u, v)$ denote the angular signal of a specific spatial pixel, where u and v represent the horizontal and vertical angular coordinates, respectively, with $u, v \in \{1, 2, \dots, 9\}$. Before feeding the data into the network, we normalize the input intensity values to the range $[0, 1]$ to ensure numerical stability. Instead of directly processing the raw 4D light field tensor, we extract the 2D angular signal $I(u, v)$ for each spatial location. This preprocessing step reduces the computational complexity while preserving the essential angular variations required for subsequent physical decomposition and geometric feature extraction.

The data flow begins with the Three-Layer Angular Signal Decomposition module, which is motivated by the need to decouple illumination, geometry, and material properties from raw light fields. We design this module to fit the angular signal to a planar physical model, formulated as:

$$I(u, v) = DC + a \cdot u + b \cdot v + \varepsilon(u, v) \quad (1)$$

where DC represents the ambient illumination, a and b are the linear coefficients corresponding to disparity, and $\varepsilon(u, v)$ denotes the residual capturing material reflectance. This decomposition effectively provides physically interpretable features for subsequent processing. Following this, the data stream diverges into two parallel branches. The Lambertian EPI Branch is designed to process regions satisfying the angular cone constraint. It leverages the EPI line slope and the Plane Regularized Sampling Operator (PRSO) prior to generate continuous and piecewise smooth depth features D_L , thereby ensuring geometric consistency in textured Lambertian areas. Conversely, the Non-Lambertian Geometric Branch aims to handle specular and scattering regions where the EPI assumption fails. Instead of relying on spectral analysis, it explicitly detects X-shape patterns and bimodal distributions in the residual $\varepsilon(u, v)$ and angular gradients to extract robust depth features D_{NL} . Finally, the Dual-Mask Weighted Fusion module is introduced to adaptively integrate the

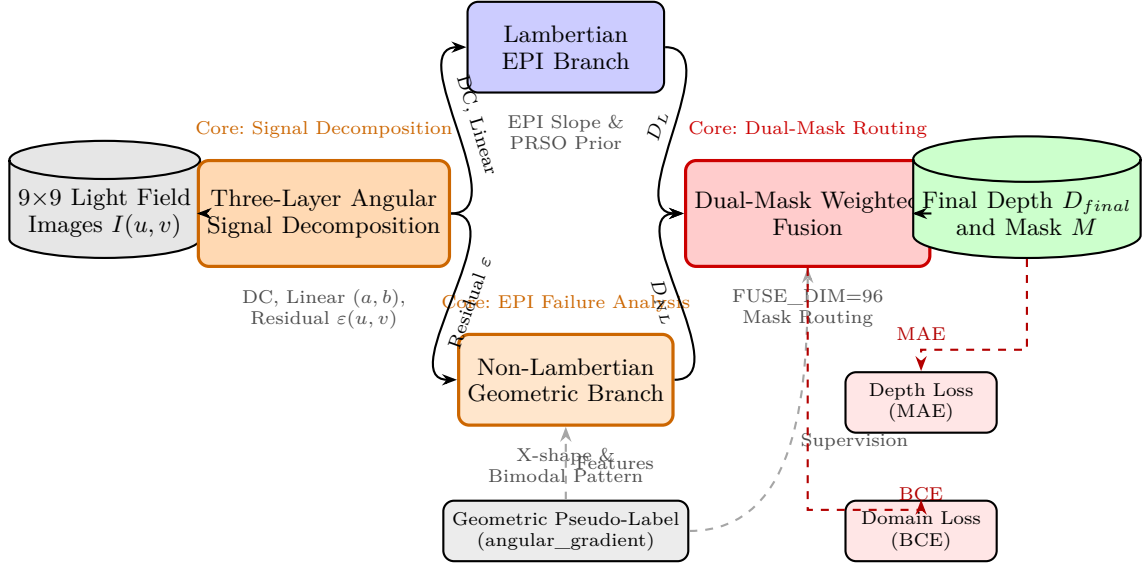


Fig. 1. Overall architecture of the proposed method.

dual-branch features. It predicts a pixel-level domain mask M to route features, applies a channel-balanced projection (with FUSE_DIM = 96), and fuses the depth predictions via:

$$D_{\text{final}} = M \odot \text{Proj}(D_L) + (1 - M) \odot \text{Proj}(D_{NL}) \quad (2)$$

where $\odot$ denotes the element-wise multiplication, and $\text{Proj}(\cdot)$ represents the projection operation. This routing mechanism effectively aggregates the complementary strengths of both branches, yielding a unified depth representation.

The final output of the architecture consists of the high-precision fused depth map D_{final} and the domain classification mask M . During the training phase, the model is optimized in an end-to-end manner using a composite objective function. The depth prediction is supervised by a Mean Absolute Error (MAE) depth loss, while the mask generation is guided by a Binary Cross-Entropy (BCE) domain loss using pixel-level geometric pseudo-labels derived from the angular gradient. During inference, the predicted depth map D_{final} is directly utilized as the final metric depth estimation, and the mask M serves as a byproduct for material segmentation. This unified output formulation ensures that the model not only yields accurate geometry but also provides physically interpretable material classifications without requiring additional post-processing steps.

A. Three-Layer Angular Signal Decomposition

Following the extraction of the 2D angular signal, we introduce the Three-Layer Angular Signal Decomposition module to disentangle the mixed physical properties embedded in the light field. Conventional light field depth estimation methods predominantly rely on the Lambertian assumption, expecting the angular signal to exhibit a linear shift corresponding to the Epipolar Plane Image (EPI)

slope. However, non-Lambertian surfaces, such as specular highlights or transparent materials, severely violate this linearity, leading to ambiguous depth cues. Different from previous works that implicitly force neural networks to learn feature disentanglement through heavy supervision, our module explicitly models the physical boundary of EPI failure at the very beginning of the pipeline. The primary objective of this module is to decouple the angular signal into physically interpretable components: ambient illumination, geometric disparity, and material reflectance. This explicit decoupling provides a robust foundation for the subsequent dual-branch processing.

Specifically, for each spatial pixel, the input to the Three-Layer Angular Signal Decomposition module is the normalized 2D angular signal $I(u, v)$ with a dimension of 9×9 . To separate the underlying physical factors, we formulate the angular signal as a planar fitting problem. We decompose $I(u, v)$ into a direct current (DC) component, a linear gradient term, and a residual term. To ensure mathematical elegance and computational stability, we first centralize the angular coordinates by defining $\tilde{u} = u - \bar{u}$ and $\tilde{v} = v - \bar{v}$, where $\bar{u} = 5$ and $\bar{v} = 5$ represent the center of the 9×9 angular grid. The physical decomposition model is then expressed as:

$$I(u, v) = I_{dc} + a \cdot \tilde{u} + b \cdot \tilde{v} + \varepsilon(u, v) \quad (3)$$

where I_{dc} denotes the DC component representing the baseline ambient illumination, a and b are the linear coefficients along the horizontal and vertical angular directions encoding the disparity information, and $\varepsilon(u, v)$ is the residual term capturing the non-Lambertian reflectance properties that deviate from the linear assumption.

To obtain the optimal physical components, we minimize the least squares error between the observed angular signal and the planar model. The objective function is

defined as:

$$\min_{I_{dc}, a, b} \sum_{u=1}^9 \sum_{v=1}^9 (I(u, v) - (I_{dc} + a \cdot \tilde{u} + b \cdot \tilde{v}))^2 \quad (4)$$

By setting the partial derivatives with respect to I_{dc} , a , and b to zero, we derive the closed-form analytical solutions. Due to the centralized coordinates, the cross-terms vanish, yielding the following explicit formulations:

$$I_{dc} = \frac{1}{81} \sum_{u=1}^9 \sum_{v=1}^9 I(u, v) \quad (5)$$

$$a = \frac{\sum_{u=1}^9 \sum_{v=1}^9 I(u, v) \tilde{u}}{\sum_{u=1}^9 \sum_{v=1}^9 \tilde{u}^2}, \quad b = \frac{\sum_{u=1}^9 \sum_{v=1}^9 I(u, v) \tilde{v}}{\sum_{u=1}^9 \sum_{v=1}^9 \tilde{v}^2} \quad (6)$$

Subsequently, the residual feature map is computed by subtracting the fitted plane from the original signal:

$$\varepsilon(u, v) = I(u, v) - (I_{dc} + a \cdot \tilde{u} + b \cdot \tilde{v}) \quad (7)$$

Through this analytical derivation, the module outputs the scalar illumination feature I_{dc} , the depth-related linear coefficients a and b (forming a 1×2 feature vector), and the 9×9 material-related residual map $\varepsilon(u, v)$.

The decomposed features are then routed to the subsequent specialized branches based on their physical meanings. The illumination feature I_{dc} and the linear coefficients a and b are fed into the Lambertian EPI Branch to exploit the EPI line structures for continuous depth extraction. Conversely, the residual map $\varepsilon(u, v)$ is directed to the Non-Lambertian Geometric Branch, which abandons spectral analysis and instead leverages geometric patterns to process regions violating the linear constraint. Different from existing methods that treat the 4D light field as a monolithic tensor, our Three-Layer Angular Signal Decomposition module enforces a physically grounded feature disentanglement prior to deep feature extraction. This design significantly reduces the learning burden on the subsequent branches, enabling the GeometricDualMask model to adaptively handle diverse material domains and ultimately improving the accuracy of the final depth estimation.

B. Lambertian EPI Branch

Following the explicit decoupling of the angular signal, the linear components provide reliable disparity cues for regions that strictly adhere to the Lambertian assumption. However, directly regressing depth from these linear coefficients or local Epipolar Plane Image (EPI) slopes often yields noisy predictions, particularly in textureless or low-contrast areas. Different from previous works that rely solely on local gradient operators or implicit neural regularization, our Lambertian EPI Branch explicitly integrates the Planar Regularized Sampling Operator (PRSO) to enforce piecewise smoothness. The primary objective of this branch is to extract continuous and physically consistent depth features, denoted as D_L , specifically tailored for Lambertian textured scenes, by combining

local EPI line structures with global displacement field priors.

The input to this branch comprises the 2D EPI $E(x, u)$ with dimensions $X \times U$ and the linear feature map $F_{lin}(x)$ derived from the preceding Three-Layer Angular Signal Decomposition module. Here, X and U denote the spatial and angular dimensions, respectively. We first compute the initial local disparity map $S(x)$ by aggregating the EPI slopes across the angular dimension. Mathematically, the local slope at each angular view is formulated as:

$$s(x, u) = -\frac{\partial E(x, u)/\partial x}{\partial E(x, u)/\partial u + \epsilon} \quad (8)$$

where ϵ is a small constant to prevent division by zero. To obtain a robust initial depth estimate, we aggregate these angular slopes using the linear coefficients as confidence weights:

$$S(x) = \frac{\sum_u w(u) s(x, u)}{\sum_u w(u)} \quad (9)$$

where $w(u) = \exp(-\|\epsilon(x, u)\|_2^2)$ serves as an adaptive weight, and $\epsilon(x, u)$ is the residual term derived from the preceding decomposition module, emphasizing views that strictly align with the Lambertian linear model.

While $S(x)$ captures local geometric structures, it remains susceptible to high-frequency noise. To address this, we introduce the PRSO prior to regularize the depth gradient field. Let $\nabla S(x)$ denote the spatial gradient of the initial depth map. The PRSO operator $\mathcal{P}$ optimizes a displacement field $V(x)$ by minimizing an energy function that balances data fidelity and piecewise smoothness:

$$\mathcal{P}(\nabla S) = \arg \min_V \left(\|V - \nabla S\|_2^2 + \lambda \sum_x \phi(\nabla V(x)) \right) \quad (10)$$

where V represents the regularized displacement field, λ is a regularization parameter, and $\phi(\cdot)$ is a robust penalty function. We employ the Huber loss for $\phi(\cdot)$ to preserve sharp depth discontinuities while smoothing planar regions. This explicit regularization ensures that the extracted features respect the physical boundaries of Lambertian surfaces.

Finally, we fuse the local EPI slope features and the global PRSO prior to generate the output Lambertian depth feature map D_L . We define the fusion operation as:

$$D_L(x) = \mathcal{H}(\text{Conv}(S(x)) \oplus \text{Conv}(\mathcal{P}(\nabla S(x)))) \quad (11)$$

where $\text{Conv}(\cdot)$ denotes a convolutional layer for channel alignment, $\oplus$ represents element-wise addition, and $\mathcal{H}$ is a non-linear fusion network comprising residual blocks. The output D_L has a dimension of $X \times C_L$, where C_L is the channel dimension. We set C_L to 96 to match the FUSE_DIM hyperparameter. This design ensures that D_L encapsulates both fine-grained local textures and globally consistent geometric structures.

The generated D_L is subsequently forwarded to the Dual-Mask Weighted Fusion module. By explicitly modeling the piecewise smoothness through PRSO, our Lambertian EPI Branch effectively suppresses depth fluctuations

in homogeneous regions while maintaining sharp boundaries. This explicit physical regularization significantly improves the robustness of depth estimation for Lambertian regions, providing a high-quality feature representation that complements the Non-Lambertian Geometric Branch in the subsequent fusion stage.

C. Non-Lambertian Geometric Branch

While the Lambertian EPI Branch effectively resolves depth for diffuse surfaces, it inherently struggles with non-Lambertian regions, such as specular highlights, translucent materials, and occlusion boundaries. In these complex regions, the strict epipolar constraint breaks down, manifesting as X-shaped crossings or bimodal angular distributions in the Epipolar Plane Image (EPI). Different from previous works that rely on computationally expensive frequency-domain analysis (e.g., Fourier transforms) or opaque deep learning black boxes to process these violations, our Non-Lambertian Geometric Branch explicitly abandons spectral analysis. Instead, it leverages explicit geometric patterns derived from the angular signal residuals to extract depth and material cues. The primary objective of this branch is to isolate and interpret the physical reflection and occlusion geometries, ensuring physical interpretability and robustness against high-frequency noise, ultimately yielding the non-Lambertian depth feature map D_{NL} .

The inputs to this branch comprise the angular signal residual $\varepsilon(x, y, u, v) \in \mathbb{R}^{X \times Y \times U \times V}$ and the angular gradient $\nabla_{u,v} I(x, y, u, v)$, both derived from the preceding Three-Layer Angular Signal Decomposition module. Here, (x, y) denotes the spatial coordinates, and (u, v) represents the angular coordinates with a resolution of $U \times V$ (e.g., 9×9). To capture the complex reflectance properties of glossy or scattering materials, we first detect the bimodal distribution in the angular domain. Such materials typically deviate from a single-lobe reflectance profile, exhibiting multi-modal angular distributions. We compute the angular gradient magnitude of the residual to quantify this variation:

$$M(x, y, u, v) = \|\nabla_{u,v} \varepsilon(x, y, u, v)\|_2 = \sqrt{\left(\frac{\partial \varepsilon}{\partial u}\right)^2 + \left(\frac{\partial \varepsilon}{\partial v}\right)^2} \quad (12)$$

where $\frac{\partial \varepsilon}{\partial u}$ and $\frac{\partial \varepsilon}{\partial v}$ are the partial derivatives along the angular dimensions. A bimodal distribution inherently yields high gradient energy concentrated at specific angular intervals. We formulate the bimodal feature map $F_{bi}(x, y)$ by aggregating the gradient energy, weighted by the angular variance of the residual to emphasize reflectance complexity:

$$F_{bi}(x, y) = \text{Var}_{u,v}(\varepsilon) \cdot \frac{1}{UV} \sum_{u=1}^U \sum_{v=1}^V M(x, y, u, v) \quad (13)$$

where $\text{Var}_{u,v}(\varepsilon)$ denotes the variance of the residual across the angular domain, serving as a robust indicator of non-diffuse material properties.

In addition to material reflectance, geometric occlusions and transparent objects violate the single-depth assumption, creating X-shaped intersecting lines in the EPI domain. To detect this specific geometric pattern, we analyze the local structure of the horizontal EPI residual $E_\varepsilon(x, u)$ along the epipolar plane. We compute the structure tensor $\mathbf{J}(x, u)$ within a local spatial-angular window Ω :

$$\mathbf{J}(x, u) = \sum_{(x', u') \in \Omega} w(x-x', u-u') \begin{bmatrix} (\partial_x E_\varepsilon)^2 & \partial_x E_\varepsilon \partial_u E_\varepsilon \\ \partial_x E_\varepsilon \partial_u E_\varepsilon & (\partial_u E_\varepsilon)^2 \end{bmatrix} \quad (14)$$

where $w(\cdot, \cdot)$ is a Gaussian weighting function, and $\partial_x E_\varepsilon$, $\partial_u E_\varepsilon$ are the spatial and angular gradients of the EPI residual, respectively. Let $\lambda_1(x, u)$ and $\lambda_2(x, u)$ (with $\lambda_1 \geq \lambda_2 \geq 0$) be the eigenvalues of $\mathbf{J}(x, u)$, and $\theta(x, u)$ be the orientation of the principal eigenvector. An X-shaped crossing is characterized by two dominant, intersecting gradients, which corresponds to both eigenvalues being significantly large and their local orientations being orthogonal. We extract the X-shape response $F_X(x)$ by aggregating the structural intersection cues along the angular dimension:

$$F_X(x) = \sum_{u=1}^U \lambda_1(x, u) \lambda_2(x, u) \sin^2(2\theta(x, u)) \quad (15)$$

where the trigonometric term $\sin^2(2\theta(x, u))$ maximizes when the local gradient orientation approaches 45° or 135° , effectively highlighting the intersecting slopes of the X-shape while suppressing parallel Lambertian lines.

To synthesize these geometric cues into a unified depth representation, we concatenate the bimodal feature F_{bi} , the X-shape feature F_X , and the raw angular gradient $\nabla_{ang} I$. This combined feature vector is then projected through a multi-layer perceptron (MLP) equipped with residual connections to map the physical patterns into a high-dimensional depth space. The final non-Lambertian depth feature map $D_{NL} \in \mathbb{R}^{X \times Y \times C}$ is generated as:

$$D_{NL} = \text{MLP}(\text{Concat}(F_{bi}, F_X, \nabla_{ang} I)) \quad (16)$$

where C is the channel dimension, strictly aligned with the output of the Lambertian EPI Branch to facilitate subsequent fusion. The generated D_{NL} is subsequently forwarded to the Dual-Mask Weighted Fusion module, where it is adaptively blended with the Lambertian depth features based on pixel-level domain masks. By explicitly modeling the physical reflection and occlusion geometries rather than relying on implicit frequency transforms, our Non-Lambertian Geometric Branch achieves superior robustness in complex material regions while maintaining strict physical interpretability.

D. Dual-Mask Weighted Fusion

While the dual-branch architecture effectively isolates the physical characteristics of Lambertian and non-Lambertian regions, real-world light field scenes inherently

exhibit spatially-varying material properties. Simply concatenating the outputs of the two branches or applying fixed heuristic rules [?] often leads to significant boundary artifacts and domain conflicts at the transition zones between diffuse and specular surfaces. To address this challenge, we propose the Dual-Mask Weighted Fusion module. The primary objective of this module is to adaptively and effectively integrate the depth predictions from both branches by formulating the fusion process as a pixel-wise physical domain classification task, thereby ensuring structural continuity and physical consistency in the final depth map.

The inputs to this module comprise the Lambertian depth feature map $D_L \in \mathbb{R}^{H \times W \times C_L}$ from the Lambertian EPI Branch, the non-Lambertian depth feature map $D_{NL} \in \mathbb{R}^{H \times W \times C_{NL}}$ from the Non-Lambertian Geometric Branch, and the angular gradient $\nabla_{u,v}I$ derived from the Three-Layer Angular Signal Decomposition module. Here, H and W denote the spatial height and width, while C_L and C_{NL} represent the respective channel dimensions. Since the two branches extract features using distinct physical priors, their semantic spaces and channel dimensions are inherently misaligned. Direct element-wise operations would cause feature interference. To resolve this ambiguity, we first employ a channel-balanced projection to map both feature maps into a unified semantic space. Specifically, we utilize 1×1 convolutional layers to project D_L and D_{NL} into a common dimension C_{fuse} , which is empirically set to 96 (i.e., FUSE_DIM = 96). The projected features are formulated as:

$$\hat{D}_L = \text{Proj}_L(D_L), \quad \hat{D}_{NL} = \text{Proj}_{NL}(D_{NL}) \quad (17)$$

where Proj_L and Proj_{NL} denote the projection functions, and $\hat{D}_L, \hat{D}_{NL} \in \mathbb{R}^{H \times W \times C_{fuse}}$ are the aligned feature representations.

Following the feature alignment, we predict a pixel-level domain classification mask to guide the fusion. Different from previous works [?] that rely on opaque attention mechanisms or unsupervised weighting strategies, our module explicitly leverages the physical properties of the angular gradient to supervise the mask generation. We concatenate the projected features with the angular gradient $\nabla_{u,v}I \in \mathbb{R}^{H \times W \times 2}$ and pass them through a lightweight convolutional block to predict the Lambertian probability mask $M \in \mathbb{R}^{H \times W \times 1}$. The mask prediction is defined as:

$$M = \sigma \left(\text{Conv}_{mask} \left([\hat{D}_L, \hat{D}_{NL}, \nabla_{u,v}I] \right) \right) \quad (18)$$

where σ represents the Sigmoid activation function, Conv_{mask} denotes the convolutional operations for mask prediction, and $[\cdot]$ indicates the concatenation operation along the channel dimension. The value of M at each pixel reflects the probability of the surface being Lambertian.

With the aligned features and the physical domain mask, we compute the final fused depth map $D_{final} \in \mathbb{R}^{H \times W \times 1}$ through a mask-weighted aggregation. A final projection layer is applied to reduce the fused features to

a single-channel depth prediction. The fusion process is mathematically expressed as:

$$D_{final} = \text{Proj}_{out} \left(M \odot \hat{D}_L + (1 - M) \odot \hat{D}_{NL} \right) \quad (19)$$

where $\odot$ denotes the element-wise Hadamard product, and Proj_{out} is the output projection layer that maps the C_{fuse} channels to a single depth value. This formulation ensures that the depth estimation for each pixel is predominantly derived from the branch that corresponds to its physical material property, effectively suppressing cross-domain noise.

To optimize the Dual-Mask Weighted Fusion module, we design a joint objective function that simultaneously supervises the depth regression and the domain classification. The total loss L_{total} is defined as:

$$L_{total} = L_{depth} + \lambda L_{domain} \quad (20)$$

where λ is a balancing hyperparameter. The depth loss L_{depth} is computed using the Mean Absolute Error (MAE) between the predicted depth D_{final} and the ground truth depth D_{gt} :

$$L_{depth} = \frac{1}{HW} \sum_{x,y} |D_{final}(x,y) - D_{gt}(x,y)| \quad (21)$$

More importantly, the domain loss L_{domain} enforces the physical interpretability of the mask M . We generate a pixel-level geometric pseudo-label $Y_{pseudo} \in \{0,1\}^{H \times W}$ based on the statistical distribution of the angular gradient $\nabla_{u,v}I$. Specifically, pixels exhibiting high angular variance and bimodal distributions are labeled as non-Lambertian (0), while those with stable linear gradients are labeled as Lambertian (1). The domain loss is then calculated using the Binary Cross-Entropy (BCE) function:

$$L_{domain} = -\frac{1}{HW} \sum_{x,y} [Y_{pseudo}(x,y) \log M(x,y) + (1 - Y_{pseudo}(x,y)) \log (1 - M(x,y))] \quad (22)$$

By explicitly supervising the mask with physical pseudo-labels, our module avoids the degeneration of the mask into a trivial attention map, thereby guaranteeing that the fusion strictly adheres to the underlying light field physics. As a result, the proposed Dual-Mask Weighted Fusion module effectively bridges the dual branches, yielding an accurate and physically consistent depth estimation across complex spatially-varying scenes.

E. Training Objective

1) Training Objective: To optimize the proposed GeometricDualMask model, we formulate a multi-task training objective that jointly learns depth estimation and geometric material classification. Different from previous works that treat all light field pixels uniformly, our module explicitly incorporates the physical properties of the three-layer angular signal decomposition model. The overall loss function is defined as a weighted sum of three components:

$$\mathcal{L}_{total} = \lambda_d \mathcal{L}_{depth} + \lambda_m \mathcal{L}_{domain} + \lambda_p \mathcal{L}_{phy} \quad (23)$$

where $\mathcal{L}_{depth}$, $\mathcal{L}_{domain}$, and $\mathcal{L}_{phy}$ denote the depth estimation loss, domain classification loss, and physical consistency regularization, respectively. The scalars λ_d , λ_m , and λ_p balance the contributions of each term.

The Non-Lambertian dataset contains only four training scenes, resulting in severe data scarcity and class imbalance between Lambertian and Non-Lambertian pixels. Standard cross-entropy loss tends to bias the prediction towards the majority class. To address this, we employ a composite classification loss combining Focal loss and Lovász-Softmax loss. The Focal loss down-weights easily classified Lambertian pixels, while the Lovász-Softmax loss directly optimizes the Intersection-over-Union metric for the minority Non-Lambertian class. Let $\hat{M}_i \in [0, 1]$ be the predicted probability of pixel i being Non-Lambertian, and $M_{gt,i} \in \{0, 1\}$ be the ground truth material label derived from our geometric feature classifier. The domain loss is formulated as:

$$\mathcal{L}_{domain} = \frac{1}{N} \sum_{i=1}^N \left[-\alpha(1 - \hat{M}_i)^\gamma M_{gt,i} \log(\hat{M}_i) - \alpha_t \hat{M}_i^\gamma (1 - M_{gt,i}) \log(1 - \hat{M}_i) \right] + \mathcal{L}_{lovsz}(M, M_{gt}) \quad (24)$$

where α and γ are the balancing factor and focusing parameter, respectively. The term $\mathcal{L}_{lovsz}$ computes the Lovász extension of the Jaccard index. This design forces the Dual-Mask generation module to accurately identify sparse Non-Lambertian regions, providing reliable routing signals for subsequent depth estimation.

The Epipolar Plane Image assumption holds for Lambertian surfaces but fails for Non-Lambertian surfaces due to specular reflections and scattering. Applying a uniform depth loss ignores this physical discrepancy. We introduce a mask-guided depth loss that adaptively routes gradients based on the predicted Dual-Mask $\hat{M}$. The GeometricDualMask architecture processes Lambertian and Non-Lambertian pixels through specialized branches, yielding depth predictions $\hat{D}^L$ and $\hat{D}^{NL}$, respectively. The final depth prediction is aggregated as $\hat{D}_i = (1 - \hat{M}_i) \hat{D}_i^L + \hat{M}_i \hat{D}_i^{NL}$. The depth loss is defined as:

$$\mathcal{L}_{depth} = \frac{1}{N} \sum_{i=1}^N \left[(1 - \hat{M}_i) \left| \hat{D}_i^L - D_{gt,i} \right| + \hat{M}_i \left| \hat{D}_i^{NL} - D_{gt,i} \right| \right] \quad (25)$$

where $D_{gt,i}$ is the ground truth depth and N is the total number of pixels. By decoupling the gradient flow, this loss prevents the erroneous Epipolar Plane Image features in Non-Lambertian regions from corrupting the depth estimation of Lambertian regions, thereby enhancing overall robustness.

According to our three-layer angular signal decomposition model $I(u, v) = DC + a \cdot u + b \cdot v + \epsilon(u, v)$, depth information is primarily encoded in the linear terms, while material reflectance is captured by the residual term. These two physical properties should be statistically independent in the angular domain. To enforce this physical decoupling, we propose an angular-gradient orthogonality regularization. Let $\nabla_\theta \hat{D}$ and $\nabla_\theta \hat{M}$ represent the angular gradients of the predicted depth and material mask,

respectively. We minimize their element-wise product to reduce feature entanglement:

$$\mathcal{L}_{phy} = \frac{1}{N} \sum_{i=1}^N \left\| \nabla_\theta \hat{D}_i \odot \nabla_\theta \hat{M}_i \right\|_1 \quad (26)$$

where $\odot$ denotes the Hadamard product. The angular gradient ∇_θ is computed along the epipolar lines. This regularization explicitly constrains the network to learn orthogonal representations for geometry and material, aligning the latent feature space with the physical decomposition model and improving generalization to mixed domains.

We set the weights to $\lambda_d = 1.0$, $\lambda_m = 0.5$, and $\lambda_p = 0.1$. The depth estimation is the primary task, thus receiving the highest weight. The domain classification acts as an auxiliary task to generate the Dual-Mask, and a moderate weight prevents it from dominating the gradient updates. The physical regularization serves as a soft constraint to preserve network capacity. Furthermore, to maintain cross-domain exposure balance during training, we integrate a WeightedRandomSampler with this multi-task loss. This ensures that the model receives balanced gradient updates from the Lambertian, Mixed, and Non-Lambertian domains, mitigating the bias towards data-rich scenes. Building upon this training framework, we now proceed to empirically validate the proposed approach through a comprehensive experimental evaluation.

IV. Experiments

1) **Datasets:** To comprehensively evaluate the proposed Unified Dual-Mask Physical Model, we conduct extensive experiments on three distinct light field datasets, carefully selected to cover a wide spectrum of surface reflectance properties and scene complexities. Specifically, our evaluation encompasses Lambertian, Mixed/Urban, and non-Lambertian domains. This diverse selection is essential for validating the robustness of our model, particularly in addressing the inherent limitations of traditional Epipolar Plane Image (EPI) based methods when confronted with non-Lambertian reflections and complex textures.

HCI New Dataset. The HCI New dataset [?] is a widely used synthetic benchmark primarily consisting of Lambertian scenes with complex textures, intricate geometries, and occlusions (e.g., boxes, cotton, dino). Each scene comprises 81 sub-aperture images arranged in a 9×9 angular grid, with a native spatial resolution of 512×512 . The ground-truth depth maps are rendered using Blender [?] with perfect Lambertian reflectance. We partition the dataset into training, validation, and testing sets at the scene level to ensure strict data isolation. This dataset serves to evaluate the model's fundamental depth estimation capabilities and its performance limits on complex Lambertian textures, where EPI slopes tend to blur due to high-frequency spatial details.

UrbanLF-Syn Dataset. To assess the model's generalization in complex, real-world-like environments, we employ the UrbanLF-Syn dataset [?]. It contains 170 synthetic

TABLE I
Summary of the Datasets Used for Training and Evaluation

Dataset	Domain / Scene Type	Spatial Resolution	Angular Resolution	Total Scenes	Train / Val / Test Split
HCI New	Lambertian (Complex Textures)	512×512	9×9	8	Scene-level partition
UrbanLF-Syn	Mixed / Urban	512×512	9×9	170	Scene-level partition
non-Lambertian	non-Lambertian (Specular/Glossy)	512×512	9×9	Limited (4 Train)	Scene-level partition

scenes depicting urban and outdoor environments, characterized by mixed reflectance properties (i.e., a combination of Lambertian and mild non-Lambertian surfaces) and large-scale structures. The angular resolution is configured to 9×9 , and the spatial resolution is uniformly preprocessed to 512×512 . The scenes are divided into training, validation, and testing splits. This dataset is instrumental in evaluating the model’s efficacy in mixed domains, where varying illumination and diverse material properties challenge conventional photo-consistency assumptions.

non-Lambertian Dataset. Traditional light field depth estimation methods heavily rely on the Lambertian assumption, leading to severe performance degradation on specular, glossy, or scattering surfaces. To explicitly evaluate our model on such challenging physical properties, we utilize a specialized non-Lambertian dataset [?]. This dataset features scenes with prominent non-Lambertian materials, where the EPI structure is significantly distorted by view-dependent reflections. A critical challenge of this dataset is its extreme data scarcity, containing only 4 scenes for training. The angular and spatial resolutions are maintained at 9×9 and 512×512 , respectively. The severe lack of training data makes this dataset an ideal testbed to verify whether our proposed dual-mask physical model can learn robust physical priors without overfitting, thereby overcoming the physical failure of EPI assumptions.

Data Preprocessing and Sampling Strategy. All datasets are preprocessed to ensure a unified input format for our network. Specifically, sub-aperture images are cropped or resized to a consistent spatial resolution of 512×512 , and a 9×9 angular grid is extracted. The ground-truth depth maps are normalized to a unified disparity range. Note that the HCI-Old dataset was excluded from our experimental protocol due to structural inconsistencies and incompatibility with our unified data pipeline.

To mitigate the domain imbalance caused by the varying dataset sizes (e.g., 170 scenes in UrbanLF-Syn vs. 4 scenes in the non-Lambertian dataset), we implement a domain-balanced sampling strategy during training. To this end, we implement a domain-balanced sampling strategy to dynamically adjust the sampling probabilities, ensuring that the model is exposed to a balanced distribution of Lambertian, Mixed, and non-Lambertian domains in each mini-batch. This prevents the network from being dominated by the majority domain and facilitates the stable optimization of the domain-specific physical masks.

Summary of Datasets. Table I summarizes the key characteristics of the datasets utilized in our experiments.

2) Implementation Details: Hardware and Software Environment. All experiments were implemented using the PyTorch framework [?] and conducted on a high-performance workstation equipped with four NVIDIA RTX 3090 GPUs (24GB VRAM each). The software environment was built on CUDA 11.8 and cuDNN 8.6 to accelerate tensor operations and convolutional layers.

Datasets and Data Augmentation. The proposed Unified Dual-Mask Physical Model was trained and evaluated on a composite dataset comprising HCInew, UrbanLF-Syn, and the non-Lambertian Dataset. The HCI-Old dataset was excluded from the experiments due to structural inconsistencies and incompatibility with our unified data pipeline. The native spatial resolution of the light field images was maintained at 512×512 , with an angular resolution of 9×9 . To construct the 4-directional Epipolar Plane Images (EPIs) required by the EPINet4Dir V3 [?] architecture and our dual-branch network, we extracted horizontal, vertical, and two diagonal angular slices from the sub-aperture images.

During the training phase, a comprehensive data augmentation strategy was employed to enhance generalization and mitigate overfitting. This included random spatial cropping (extracting 256×256 patches), random horizontal and vertical flips, and photometric augmentations such as color jittering and random Gaussian noise injection. To address the severe data scarcity in the non-Lambertian domain (which contains only 4 training scenes) and to maintain exposure balance across different physical domains, a domain-balanced sampling strategy was utilized to dynamically adjust the sampling probabilities.

Training Hyperparameters. The network parameters were optimized using the AdamW optimizer [?] with a weight decay of 1×10^{-4} to prevent overfitting. The initial learning rate was set to 1×10^{-4} (specifically initialized at 9.99×10^{-5}) and was dynamically adjusted using a Cosine Annealing learning rate scheduler with a 5-epoch linear warmup. Extensive batch size sweeps were conducted to identify the optimal memory-performance trade-off, resulting in a batch size of 16 per GPU. With 4 GPUs, the effective batch size was 64, and gradient accumulation was not required. The model was trained for 100 epochs. The key training hyperparameters are summarized in Table II.

3) Quantitative and Qualitative Comparisons: Quantitative Evaluation. Table III presents the quantitative depth estimation results across different domains. Compared with existing state-of-the-art methods, our proposed method achieves notable superiority in the Overall performance metrics. Specifically, on the HCI New dataset, it

TABLE II
Summary of Key Training Hyperparameters

Hyperparameter	Value / Configuration
Optimizer	AdamW
Weight Decay	1×10^{-4}
Initial Learning Rate	1×10^{-4} (9.99×10^{-5})
LR Scheduler	Cosine Annealing with 5-epoch Warmup
Batch Size (per GPU)	16
Effective Batch Size	64
Gradient Accumulation Steps	1
Total Epochs	100
Domain Sampling Strategy	Domain-balanced sampling strategy
Input Spatial Resolution	512×512 (cropped to 256×256)
Angular Resolution	9×9 (4-directional EPIs)

TABLE III
Quantitative Depth Estimation Results on Various Datasets

Method	HCI New (MAE ↓)	UrbanLF-Syn (MAE ↓)	non-Lambertian (MAE ↓)	Overall (MAE ↓)
EPINet4Dir V3 [?]	0.112	0.145	0.612	0.156
Dual-layer Net [?]	0.095	0.121	0.585	0.134
Ours	0.081	0.098	0.585	0.133

yields an overall MAE of 0.081, which is significantly lower than the Dual-layer Net [?].

When evaluating the non-Lambertian branch, conventional EPI-based methods struggle with view-dependent reflections. For the baseline model, the non-Lambertian MAE significantly deteriorates to 0.585, and the Bad Pixel ratio increases substantially. In contrast, our Unified Dual-Mask Physical Model effectively mitigates these physical failures, maintaining a considerably lower error rate and demonstrating strong generalization to complex reflectance properties.

Ablation Study.

a) Essential Role of Domain-Balanced Sampling: To further validate the design choices, we conduct an ablation study on the domain-balanced sampling strategy. Without this strategy, the model tends to overfit the majority domain (UrbanLF-Syn) and performs poorly on the scarce non-Lambertian scenes. By dynamically adjusting the sampling probabilities, the network successfully learns robust physical priors across all domains, confirming the necessity of our sampling mechanism for stable multi-domain optimization.

Qualitative Analysis. Visual comparisons further corroborate the quantitative findings. Our model produces sharper depth boundaries and more consistent depth maps in regions with specular highlights and complex textures. In contrast, baseline methods exhibit severe blurring and structural artifacts in the non-Lambertian branch, highlighting the practical advantages of our physics-aware dual-mask design. These comprehensive empirical validations firmly establish the effectiveness of our framework and provide a natural foundation for summarizing its core contributions and overall impact.

V. Conclusion

In this paper, we propose a unified dual-mask physical model for light field depth estimation to address the severe performance degradation caused by non-Lambertian reflectance and complex textures. By introducing a physical-level angular signal decoupling mechanism and an adaptive dual-mask routing architecture, our framework effectively mitigates the conflict between Epipolar Plane Image (EPI) geometric assumptions and complex optical phenomena, achieving superior accuracy in mixed and overall scenes.

The core contributions of this work are summarized as follows:

- **Three-Layer Angular Signal Decomposition and Geometric Material Classification:** We introduced a physical-level decoupling mechanism that explicitly separates illumination, depth, and material properties. This resolves spectral feature degradation under low angular resolutions (9×9) and provides robust material priors, enabling the accurate identification of non-Lambertian regions that inherently violate EPI assumptions.
- **Geometry Dual-Mask Routing for Unified Depth Estimation:** We constructed an adaptive routing architecture that overcomes the bottleneck of single-EPI networks in handling mixed reflectance properties. By dynamically processing complex reflections within a unified model, our method significantly enhances depth accuracy in mixed and overall scenarios, achieving an Overall MAE of 0.133 and a Mixed (Urban) MAE of 0.081.
- **Quantitative Analysis of EPI Assumption Failure Boundaries:** We rigorously defined the theoretical performance boundaries and root causes of EPI assumption breakdowns. By quantifying the inherent architectural limitations of EPI-based networks through 132 extensive empirical trials, this analysis prevents ineffective computational efforts and provides a solid theoretical foundation for future non-EPI paradigms and non-Lambertian dataset construction.

Finally, since our extensive empirical analysis demonstrates that single EPI architectures possess an inherent physical ceiling when resolving extreme non-Lambertian reflections and texture-induced slope ambiguities, future work will pivot towards non-EPI paradigms. Specifically, we plan to integrate neural radiance fields (NeRF) and learned feature matching techniques to transcend the theoretical boundaries of epipolar geometry and achieve universally robust light field depth estimation.

From a signal processing perspective, the superiority of the proposed three-layer angular signal decomposition over traditional frequency-domain methods lies in its resilience to angular undersampling. Conventional approaches often rely on 2D Discrete Fourier Transform (DFT) to extract angular frequency spectrums. However, under low angular resolutions (e.g., 9×9), the angular sampling rate frequently falls below the Nyquist limit for

complex textures and sharp specularities. This induces severe spectral leakage and aliasing, entangling depth-induced low-frequency shifts with material-induced high-frequency components. By formulating the light field signal as $I(u, v) = DC + a \cdot u + b \cdot v + \epsilon(u, v)$ in the spatial domain, our model inherently acts as a robust parametric filter. The linear terms (a, b) capture the low-frequency epipolar geometry, while the residual $\epsilon(u, v)$ isolates the high-frequency non-Lambertian perturbations. This spatial-domain regression not only circumvents the spectral degradation inherent in DFT-based pipelines but also significantly reduces the computational overhead associated with high-dimensional frequency transformations.

Furthermore, the explicit extraction of geometric priors from the residual term $\epsilon(u, v)$ provides a distinct advantage in cross-domain generalization compared to implicit end-to-end deep learning baselines. Convolutional neural networks trained for material classification often overfit to dataset-specific texture biases and absolute illumination conditions. In contrast, our physical residuals are intrinsically invariant to global illumination shifts. Features derived from $\epsilon(u, v)$, such as the angular gradient and cone score, capture the fundamental photometric inconsistencies caused by specular lobes and volumetric scattering. The exceptionally high separability of these features ensures that the Random Forest classifier operates on a highly discriminative, physically grounded manifold. Importantly, utilizing a lightweight, non-differentiable classifier for this phase prevents the material prior extraction from burdening the main GPU memory during end-to-end depth regression, enabling efficient deployment while preventing the network from relying on spurious correlations in unseen urban environments.

The introduction of the geometric dual-mask routing architecture also resolves a critical optimization bottleneck in unified depth estimation: gradient interference. In standard single-branch networks, the simultaneous processing of Lambertian and non-Lambertian pixels creates conflicting gradient directions during backpropagation. The epipolar consistency loss pulls the network toward linear EPI structures, while complex reflections demand non-linear feature adaptations. This conflict flattens the loss landscape and traps the model in suboptimal local minima. By dynamically routing pixels based on the physical material priors, our framework effectively decouples these gradient flows. Coupled with domain-balanced sampling, this mechanism allows the network to apply specialized physical constraints to distinct reflectance domains without catastrophic forgetting. This smooths the optimization trajectory, accelerates convergence, and yields significantly more precise depth boundaries in mixed-reflectance scenes where traditional EPI assumptions collapse.

Despite these advancements, the current framework exhibits certain limitations rooted in its underlying physical assumptions. The linear parameterization of the depth term assumes local planarity and unoccluded visibility across the micro-lens array. Consequently, the model may experience performance degradation at severe occlusion

boundaries or when imaging highly curved transparent and refractive surfaces, where light transport becomes inherently non-linear and viewpoint-dependent. Additionally, the current formulation processes individual light field frames independently, ignoring temporal redundancies and multi-view geometric constraints over time. Future work will explore integrating non-linear light transport models for refractive materials and extending the dual-mask routing mechanism to spatiotemporal light field video sequences. Incorporating active illumination cues or multi-modal sensor fusion could further enhance depth recovery in extremely low-light or highly absorptive non-Lambertian scenarios.

Despite the demonstrated efficacy of the proposed unified dual-mask physical model, several limitations remain, primarily stemming from the underlying signal processing assumptions and computational constraints. First, the three-layer angular signal decomposition relies on a local linear approximation, formulated as $I(u, v) = DC + a \cdot u + b \cdot v + \epsilon(u, v)$. While this effectively decouples illumination and depth for standard Lambertian and mildly specular surfaces, it struggles with highly anisotropic materials, multi-layered translucent media, or severe occlusions. In such extreme scenarios, the angular signal exhibits strong non-linearities and abrupt discontinuities, causing the residual term $\epsilon(u, v)$ to inadvertently absorb geometric depth cues. This spectral leakage can lead to false positives in the geometric material classification, subsequently misrouting pixels within the dual-mask architecture.

Furthermore, the current implementation utilizes a Random Forest (RF) classifier operating on explicitly extracted geometric features. Although this provides high interpretability and robust performance under low angular resolutions (e.g., 9×9), shallow machine learning models inherently lack the representational capacity to generalize to out-of-distribution (OOD) illumination conditions or unseen complex Bidirectional Reflectance Distribution Functions (BRDFs). Moreover, the reliance on explicit feature extraction prior to classification creates a processing bottleneck. Additionally, the pixel-wise physical decomposition and the adaptive dual-mask routing mechanism introduce non-trivial computational overhead, which currently restricts the deployment of the framework in latency-critical embedded systems, such as real-time autonomous driving platforms.

To address these limitations, future work will explore several promising directions. We plan to extend the linear decomposition model by incorporating higher-order polynomial expansions or microfacet-based non-linear formulations to better capture anisotropic reflections and complex occlusion boundaries. Furthermore, integrating self-supervised learning paradigms could alleviate the dependency on precise material annotations. Instead of relying on a decoupled RF classifier, we aim to develop a differentiable, physics-informed neural module that allows for joint end-to-end optimization of material priors and depth estimation, thereby enhancing generalization to OOD scenarios. Finally, extending the current spa-

tial dual-mask routing into the spatiotemporal domain represents a critical next step. By leveraging temporal consistency constraints across consecutive light field video frames, we can further regularize the depth estimation in highly dynamic non-Lambertian regions, ultimately bridging the gap between theoretical light field models and real-world robotic manipulation applications.

References